

Sample Centrifugal Pump Maintenance Manual

Equipment: P-101 Centrifugal Process Pump

Purpose: Test document for an industrial RAG / agent framework. This PDF is intentionally created without password protection or AES encryption.

1. Safety

Stop the pump, isolate electrical power, close process valves when required, and follow the site's lockout/tagout procedure before maintenance. Allow hot equipment to cool before inspection.

2. Normal Operating Checks

During operation, monitor suction pressure, discharge pressure, flow rate, motor current, bearing temperature, vibration, and unusual noise. Record readings so changes can be compared over time.

3. Low Discharge Pressure

Possible causes of low discharge pressure include a blocked suction strainer, insufficient liquid level at the suction source, air entering the suction line, incorrect pump rotation, excessive system demand, worn impeller surfaces, or a partially closed suction valve.

Recommended checks: verify suction valve position, inspect and clean the suction strainer, check for suction-line leaks, confirm the pump is fully primed, verify motor rotation, and inspect the impeller if the problem continues.

4. Low Flow

Low flow can occur because of suction restrictions, discharge restrictions, air pockets, low pump speed, an obstructed impeller, or operation far from the pump's intended duty point. Compare measured suction and discharge pressures with normal baseline values.

5. Excessive Vibration

Excessive vibration may indicate pump and motor misalignment, loose mounting bolts, worn bearings, cavitation, an unbalanced or damaged impeller, or pipe strain. Do not continue operating equipment with rapidly increasing vibration.

6. Cavitation

Cavitation can produce a crackling or gravel-like sound and may reduce flow and discharge pressure. Check for restricted suction piping, clogged strainers, low liquid level, excessive liquid temperature, or other conditions that reduce available suction pressure.

7. High Motor Current

High motor current may be associated with excessive flow, mechanical binding, bearing problems, incorrect voltage, or pump operation outside the intended range. Compare current against the motor nameplate rating and the normal operating baseline.

8. Preventive Maintenance

Interval	Inspection
Daily	Check pressure, flow, vibration, noise, leakage, and motor current.
Weekly	Inspect visible fasteners, coupling guard, and seal area.
Monthly	Review trends and inspect suction strainer condition where applicable.
Quarterly	Check alignment, mounting condition, and bearing condition.
Annually	Perform detailed inspection based on operating hours and service conditions.

9. Troubleshooting Example

Symptom: P-101 has lower-than-normal discharge pressure. First verify that the suction valve is open and the suction strainer is clean. Next check whether air is entering the suction piping and confirm that the pump is primed. If these conditions are normal, verify rotation and inspect the impeller for wear or blockage.

10. RAG Test Questions

Try asking your agent: 'What can cause low pump pressure?', 'What should I inspect when P-101 vibrates?', 'What are signs of cavitation?', or 'What should be checked when motor current is high?'

Note: This is synthetic demonstration content for software testing, not a manufacturer-specific maintenance procedure.